MILITARY SPECIFICATION

VOLTAGE REGULATOR SUBASSEMBLY: 10525755

This amendment forms a part of Military Specification MIL-V-60217(MU), dated 2 March 1965.

Table I: Add the following note below Table I:

"NOTE: Resistor R8 plus R9 shall equal 30.0 ohms or 30.1 ohms ± 0.5
Resistor R60 plus R61 shall equal 70 ohms ± 0.5 percent.
Resistor R43 plus R44 shall equal 110 ohms ± 0.5 percent.
Resistor R67 plus R68 shall equal 110 ohms ± 0.5 percent.
Resistor R71 plus R72 shall equal 110 ohms ± 0.5 percent."

Table II, under Voltage Inputs: Delete the columns and substitute the following columns:

"NAME	LOW LINE (volts)	NOMINAL LINE (volts)	HIGH LINE (volts)
+150v main	+193.0 ± 6.0	+232.0 ± 6.0	+274.0 ± 6.0
+150V bias	+193.0 ± 6.0	+232.0 ± 6.0	+274.0 ± 6.0
+20v	+16.0 ± 1.0	+19.5 ± 1.0	+23.0 ± 1.0
+15v main	+21.0 ± 1.2	+26.0 ± 1.5	+30.8 ± 1.2
+6v	+5.5 ± 0.05	+6.0 ± 0.05	+6.5 ± 0.05
-12v	-12.0 ± 1.2	-12.0 ± 1.2	-12.0 ± 1.2
-27v	-21.0 ± 1.0	-25.5 ± 1.0	-29.6 ± 1.6
-32v	-26.0	-31.0	-37.0

Table II, under Trip Load (amperes), lines 1 and 2: Delete "0.19 ± 0.02" and "1.7 ± 0.1" and substitute "0.19 $^{+0.04}_{-0.0}$" and "1.7 $^{+0.2}_{-0.0}$"; add new column under "Loads".

"EXTREME TEMPERATURES TRIP LOAD (amperes)
$0.19\ ^{+0.06}_{-0.02}$
$1.7\ ^{+0.3}_{-0.1}$"

Following paragraph 4.1.1, add paragraphs 4.1.2, 4.1.2.1 and 4.1.2.2.

FSC 1220

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"4.1.2 First Article (Initial Production) Approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) voltage regulators manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.1.2.1 Government Testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor Testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

Paragraph 4.2.3: Add the following sentence: "No lots shall be accepted until control samples have been satisfactorily tested."

Paragraph 4.3.2: Delete first sentence and substitute:

"Voltage regulator units shall be selected at random by a Government representative as control samples from each three (3) months production or portion thereof as follows:

3 units per first 100 produced
5 units per 200
7 units per 300
9 units per 400
10 units per 500
12 units per 700
15 units per 1000
and 2 units for each 100 items produced in excess of 1000."

Paragraph 4.5.1.3.2: Delete in its entirety and substitute:

"4.5.1.3.2 Operating.- This test shall be conducted in accordance with the humidity test defined in Procedure III of MIL-F-13926, except that the humidity shall be maintained at 92 to 97 percent for the 24-hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test period, the board shall be

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capable of performing the tests of 4.5.8. In order to demonstrate this capability, the tests of 4.5.8 shall be performed a minimum of once each 6 hours of the 24-hour period. The final test shall be conducted during the final portion of the 24-hour period. Between tests, the board shall be electrically de-energized."

Paragraph 4.5.1.4: Delete in its entirety and substitute:

"4.5.1.4 Salt spray.- This test shall be conducted in accordance with the salt spray test defined in Method 101, Condition B of MIL-STD-202 using a 5 percent salt solution. At the end of the exposure period, salt deposits shall be removed from the board by a gentle wash, not exceeding 30 minutes, in running water not warmer than 100°F, using a soft-hair brush or plastic-bristle brush. The board shall then be air dried in a controlled environment at 72°F + 5°F and a relative humidity of 30 to 50 percent for a period of 24 hours. The board shall then be inspected visually for any physical damage, then tested in accordance with 4.5.4 to 4.5.8.6 inclusive."

Paragraph 4.5.3.5: Add to end of paragraph "except for the trip current resistors. The trip resistor values shall be those determined by selecting the ratio of resistors (R68 to R67) for +150 volts and (R43 to R44) for +15 volts that causes tripping at a current nearest to but ≥ .19 amperes for +150 volt trip and nearest to but ≥ 1.7 amperes for the +15 volt trip."

Paragraph 4.5.8.1(b): After "150V main" add "plus 150V bias".

Paragraph 4.5.8.3: Delete "0.19 amperes" and substitute "0.23 amperes".

After paragraph "4.5.8.6" add new paragraph:

"4.5.9 Extreme environmental conditions.- All tests of 4.5.8 shall be made using the extreme temperature columns of Table II where applicable. A minimum of 100 feet per minute of air at ambient temperature shall flow across the components mounted on the board."

Table VI, Three Phase Full Wave Bridge Types*: Delete second column and substitute new second column.

"VARIABLE RANGE OF OUTPUT (volts)
+187 to +280
+187 to +280
+ 19 to + 32
+ 15 to + 24
- 20 to -31.2"

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Figure 6, below "-32V P.S.": Add phrase "see Figure 5".

* The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous amendment.

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